

GUJARAT TECHNOLOGICAL UNIVERSITY
GRADUATE SCHOOL OF ENGINEERING AND TECHNOLOGY

Course Laboratory Experiment List (INDEX)

CourseCode & Name		BE05000231_PDS					
Academic Term:		AY-2026-27-ODD Term			Semester		V
Student Enrollment No:					Batch:		
Student Name:							
Sr No	Objective/Aim	CO /PO / PSO	Date of Assigned	Date of Completion	Marks	Sign	
1	To Load and Explore the Unstructured Access Log Data Objective: Learn how to import and explore raw server access logs (Apache/Nginx format). ● Load a sample access.log file using Python (e.g., with open() or pandas + regex). ● Understand the typical log structure (IP, timestamp, request, status, size, user-agent,etc.). ● Extract 5–10 random lines to manually inspect the pattern. ● Identify useful fields for analysis.	CO1	06.07.2026				
2	To Convert the Unstructured Log Data into a Structured Dataset Objective: Parse log data into structured tabular form. ● Use regex or apachelogs, pyparsing, or logparser libraries to extract fields: ○ IP address ○ Date/time ○ Request type (GET, POST) ○ Status code ○ Resource requested ○ Bytes sent ○ Referrer ○ User-agent ● Store data in a Pandas DataFrame	CO1	20.07.2026				

3	To clean the data and Preprocessing for further process of dataset. Objective: Clean structured data for analysis. • Convert timestamp to datetime format. • Handle missing values and inconsistent entries. • Normalize URL paths (e.g., /index.html vs /index). • Lowercase strings and remove extraneous spaces.	C02	03.08.2026			
4	To label the requests as Benign or Attack (Basic Classification) from the dataset. Objective: Create labeled categories for analysis. • Use known attack patterns (e.g., SQL injection, path traversal, brute-force login) to label logs. • Example: ◦ SQLi: URLs with ' OR 1=1 -- ◦ Brute force: Repeated login attempts from same IP ◦ Path traversal: ../ in the URL • Create a new column: label → benign, sqli, brute_force, etc	C02	17.08.2026			
	To do the feature engineering for the anomaly detection or any suspicious activity. Objective: Create meaningful features for machine learning. • Number of requests per IP • Time between requests • Status code frequency (e.g., many 404 → suspicious) • User-agent parsing (bots vs real browsers) • URL entropy (suspicious randomness)	C03	31.08.2026			
	To do the balancing of the Dataset for ML training/Deep Learning training. Objective: Handle imbalanced dataset for fair ML training. • Analyze distribution of benign vs attack traffic. • Use techniques like: ◦ Undersampling of benign ◦ Oversampling of attack (SMOTE or random oversampling) • Prepare a balanced dataset.	C03	07.09.2026			
	To do the Data Wrangling for Aggregated Analysis on the balanced dataset. Objective: Aggregate and reshape data for deeper insights. • Group by IP to analyze frequency of attacks • Time-series resampling (e.g., hourly/daily traffic) • Use pivot tables to see request type by label • Filter out bots and internal IPs	C04	21.09.2026			

To use the visual tools or by using python for the Data Visualization and EDA Objective: Use visual tools to understand the dataset. ● Plot: ○ Requests per hour ○ Top 10 attacking IPs ○ Attack categories over time ○ Status code distributions ○ Heatmaps of IP vs request types ● Tools: Matplotlib, Seaborn, Plotly Practical 9: To build a Simple Classifi	C04	05.10.2026			
To build a Simple Classifier to detect the attacks. Objective: Train a basic model to detect attacks. ● Use features from practical 5 ● Split into train/test ● Apply Logistic Regression or Random Forest ● Evaluate accuracy, confusion matrix	C05	12.10.2026			
To create a Reusable Data Pipeline for the log file. Objective: Wrap the entire log processing into a pipeline. ● Automate: ○ Loading logs ○ Parsing & structuring ○ Labeling ○ Preprocessing ○ Feature engineering	C05	12.10.2026			

Date of File Submission:_____

Course Coordinator:_____

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AY-2026-27-ODD Term

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PRACTICAL - 1

CourseCode & Name	BE05000231_PDS		
Academic Term:	AY-2026-27-ODD Term	Semester	V
Student Enrollment No:		Batch:	
Student Name:			

AIM/Objective:	
1	

Expected Outcome:		CO/PO/PSO
1		

Student Report with Experiment Results and Analysis

Experimental report will include:

- Resources Used, Experimental Setup, Program Code etc
- Analysis on results
- Concluding Remarks & Findings

Evaluation Rubrics		Marks	Inadequate	Good	Excellent
			0%	50%	100%
1	The understanding of the Student regarding the objective of the given practical	2			
2	Installation of Software or Hardware Setup level	2			
3	Quality of the Analysis done	2			
4	Quality of the report including concluding remarks and Findings	2			
5	Question & Answer related to given practical & timely submission	2			
		10			
Total Marks Obtained Out of 10					

Date of Completion:_____

Course Cordinator:_____